Smart Pest Management System

Features and Technology Stack Summary

1. Key Features

Farmer Interface

- AI Disease Detection: Upload leaf images to detect diseases offline (MobileNetV2).
- Medicine Recommendations: Get Organic, Inorganic, and Homemade treatment options.
- Offline Chatbot: Ask questions about crops and diseases without internet.
- Shop Locator: Find nearby pesticide shops.
- Forums: Discuss issues with other farmers (Community Meds).
- Government Schemes: View latest agricultural subsidies.

Admin Interface

- User Management: Manage Farmers and Experts (Suspend/Delete).
- Inventory Control: Manage Shops and Medicines.
- Visual Filtering: Color-coded medicine filters (Organic=Green, Inorganic=Red).
- AI Model Retraining: Update the detection model directly from the dashboard.
- Analytics: View system usage stats and activity logs.

Expert Interface

- Consultation Dashboard: View and reply to farmer consultation requests.
- Case History: Track solved cases.

2. Technology Stack

Frontend (User Interface)

- React.js: Component-based UI framework.
- Vite: Fast build tool and dev server.
- Tailwind CSS: Utility-first styling for a responsive, nature-themed design.
- Lucide React: Modern icon set.

Backend (Server & logic)

- Python Flask: Lightweight WSGI web application framework.
- PyTorch: Deep learning framework for the CNN model.
- Pillow (PIL): Image processing.

Database

- MongoDB: NoSQL database for flexible data storage (Users, Medicines, Logs).
- PyMongo: Python driver for MongoDB.

AI & Machine Learning

- MobileNetV2: Efficient Convolutional Neural Network (CNN) optimized for local deployment.
- Torchvision: Transformations and model architecture.

Deployment & Environment

- Offline-First Architecture: Core AI runs locally.
- REST API: Communication between frontend and backend.